

Leakage-Safe Multi-Seed Evaluation of Published BU-Net Components for Resource-Constrained 2D Glioma Segmentation

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Running title: Controlled evaluation of BU-Net components

Article type: Original research - methodological evaluation

Abstract

Background: Architectural comparisons in glioma segmentation are vulnerable to patient leakage, inconsistent training budgets, incomplete attribution, and selective reporting. We evaluated previously published BU-Net components under a single guarded protocol rather than proposing a new architecture.

Methods: BraTS 2020 training cases formed the canonical labeled cohort. BraTS 2019 was used only to audit identity overlap. All 369 canonical patients were split at patient level into 258 training, 37 validation, and 74 internal held-out test cases. A standard 2D U-Net, the published BU-Net reimplementations with residual extended skip (RES) and wide context (WC) modules, and U-Net+RES were trained with identical preprocessing, loss, optimization, and 2,000-step per-run limits. Development screening preceded a five-seed finalist stage. The test manifest was opened once after candidate and analysis freezing. The primary outcome was patient-level mean Dice across whole tumor, tumor core, and enhancing tumor. Three paired contrasts used 10,000 bootstrap resamples, 100,000 sign-flip permutations, and Holm correction.

Results: On the internal held-out test subset, mean regional Dice was 0.736 for standard U-Net, 0.752 for BU-Net, and 0.756 for U-Net+RES. Relative to standard U-Net, paired differences were +0.017 for BU-Net (95% CI +0.005 to +0.030; Holm $p=0.014$) and +0.020 for U-Net+RES (95% CI +0.008 to +0.034; Holm $p=0.004$). BU-Net was lower than U-Net+RES by 0.004 (95% CI -0.008 to -0.000; Holm $p=0.050$). BU-Net used 8.385 million parameters versus 4.450 million for U-Net+RES and had higher median latency and peak allocated memory.

Conclusions: Under this bounded 2D protocol, the published RES component was associated with a small improvement over standard U-Net. Adding WC in the full BU-Net did not improve the primary endpoint over RES alone and increased resource demand. These internal, single-dataset findings do not establish clinical utility, external generalization, or superiority over 3D, transformer, or self-configuring systems.

Keywords: brain tumor segmentation; BraTS; U-Net; BU-Net; reproducibility; patient-level evaluation; resource profiling

1. Introduction

Glioma segmentation supports quantitative analysis of multimodal magnetic resonance imaging (MRI), but model rankings can be distorted by experimental design. Slice-level partitioning can place images from one patient in multiple subsets. Reusing overlapping BraTS editions can duplicate subjects. A comparison can also favor one architecture through a different loss, augmentation policy, run duration, or stopping rule. Finally, an overlap score alone does not describe boundary error, lesion detection, failure modes, or computational demand.

The BraTS benchmark standardized multimodal glioma MRI and the evaluation of whole tumor (WT), tumor core (TC), and enhancing tumor (ET) [1-4]. U-Net remains a useful reference architecture [5]. Rehman et al. later described BU-Net, a 2D U-Net modification containing residual extended skip (RES) and wide context (WC) modules [6]. These modules are prior work; we do not claim either component as novel. Our question is narrower: what evidence do RES and WC provide when implemented in one codebase and evaluated with matched data, training, statistical, and resource protocols?

Contemporary segmentation systems include self-configuring pipelines and volumetric transformer designs [9,14,15]. They are important context, but they were not trained here. Comparing our internal scores with literature values produced under different cohorts and protocols would not be a controlled benchmark. We therefore position this work as a component and reproducibility study within a 2D U-Net family.

The study makes four contributions:

1. a content-based audit of the overlap between BraTS 2019 and BraTS 2020 before selecting one canonical cohort;
2. a patient-level development and test workflow with a frozen, one-time internal-test access event;
3. a multi-seed, matched-protocol evaluation of standard U-Net, U-Net+RES, and the full BU-Net reimplementation; and
4. artifact-derived statistical, resource, subgroup, and qualitative reports released with executable checks.

No contribution is framed as a new network block, clinical device, or state-of-the-art result.

2. Related work

BraTS combines T1, post-contrast T1, T2, and FLAIR MRI with expert tumor annotations [1-4]. The official tasks define WT as labels 1, 2, and 4; TC as labels 1 and 4; and ET as label 4 [2]. Dice and 95th-percentile Hausdorff distance (HD95) are established challenge measures, but each captures a different property of the segmentation.

U-Net introduced an encoder-decoder design with skip connections for biomedical segmentation [5]. BU-Net retained the 2D U-Net form while adding RES modules to the skip pathways and a WC module at the bottleneck [6]. The large separable kernels in these modules aim to increase contextual support. Because RES and WC originated in BU-Net, the present implementation and all component names are attributed to Rehman et al.

Loss design also matters for imbalanced lesions. Tversky and focal Tversky objectives modify the relative weighting of false-positive and false-negative errors [7,8]. We screened these objectives under the same development protocol rather than assigning different losses to different final models.

Methodological guidance recommends choosing metrics for the task, preserving the correct unit of analysis, reporting the split level, declaring primary outcomes, and making software available [10,12]. We therefore used the patient as the statistical unit, treated secondary endpoints as estimation-only analyses, and retained

undefined or infinite outcomes instead of silently replacing them. Surface Dice was included at a predeclared 1 mm tolerance [11]. That tolerance is an analytic setting, not a claim of clinical acceptability or interobserver calibration.

3. Materials and methods

3.1 Study design and reporting scope

This was a retrospective computational study of a public, de-identified challenge dataset. The workflow was organized into sequential gates: data integrity and identity audit; patient-level split; preprocessing and evaluator validation; model and loss implementation tests; single-seed development screening; multi-seed confirmation; analysis freezing; one guarded internal-test evaluation; and artifact-derived reporting. CLAIM 2024 informed the reporting checklist [12].

The study did not use the official BraTS 2020 validation set because its reference labels were withheld. In this paper, “internal held-out test subset” refers only to the labeled partition created from the BraTS 2020 training cohort.

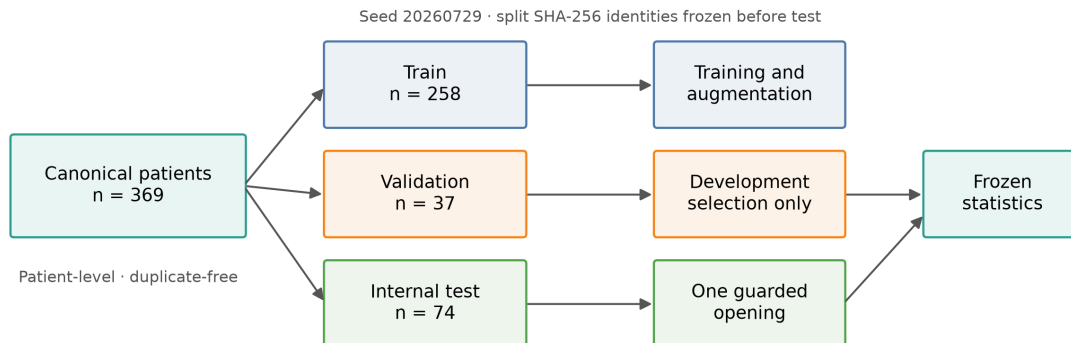


Figure 1: Gated development, freezing, and one-time internal-test analysis flow.

3.2 Cohort audit, canonicalization, and splitting

We inventoried all expected modalities and segmentations by patient and computed file- and image-content signatures. BraTS 2020 training was selected as the canonical labeled cohort. BraTS 2019 contributed no additional training observation and was used only for identity and duplicate auditing.

The audit found 335 complete BraTS 2019 cases and 369 complete BraTS 2020 cases. All 335 BraTS 2019 patients mapped to BraTS 2020 by image content; 34 cases were new in the 2020 edition. The mapped image modalities were content-equivalent. One mapped segmentation differed by 2 voxels, which was recorded as an annotation revision rather than a new patient. The canonical cohort therefore contained 369 unique patients.

Patients, not slices, were partitioned with seed 20260729. The selected candidate minimized imbalance across grade, ET presence, and WT-volume strata while satisfying frozen tolerances. Patient identifiers did not overlap between subsets.

Table 1. Cohort characteristics by patient-level partition.

Partition	Patients	HGG	LGG	ET present	WT volume median (IQR), mm3
Train	258	204	54	239	90936.0 (50321.0- 144849.5)
Validation	37	30	7	34	89516.0 (52898.0- 149466.0)
Internal test	74	59	15	69	89554.0 (51805.8- 141632.2)

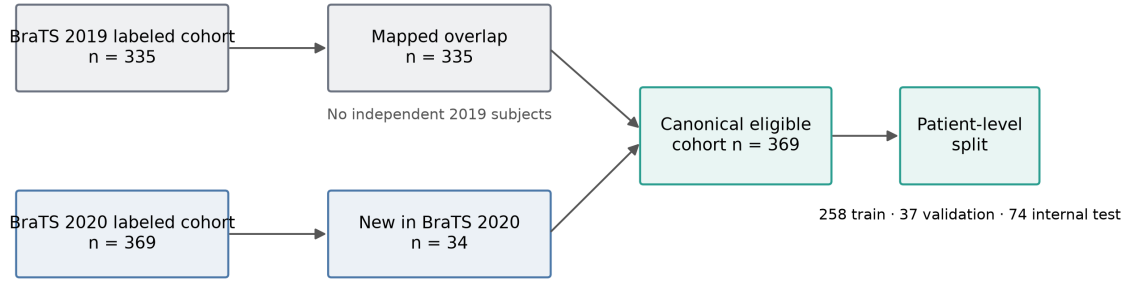


Figure 2: Cohort identity audit and canonical patient flow.

3.3 MRI preprocessing and sampling

Each case used T1, T1ce, T2, and FLAIR in that channel order. Volumes supplied by BraTS were co-registered, skull stripped, and sampled axially at 240 x 240 pixels. For each patient and modality, nonzero voxels were standardized with that volume’s mean and standard deviation. Intensity clipping was disabled.

Training sampled 16 slices per patient per epoch; the probability of selecting a tumor-containing slice was 0.67 and at least one tumor voxel defined a positive slice. Spatial augmentation used independent flips with probability 0.5 and rotations by multiples of 90 degrees. Per-modality intensity augmentation, applied with probability 0.5, used scale 0.9-1.1 and shift -0.1 to 0.1 in standardized units. Validation and test evaluation traversed all slices, including empty slices, deterministically. Cached arrays were memory-mapped outside the raw-data roots.

3.4 Architectures and attribution

All candidates accepted four MRI channels and produced four logits representing background and BraTS labels 1, 2, and 4. The common encoder widths were 16, 32, 64, and 128, followed by a 256-channel bottleneck. Batch normalization and dropout probability 0.3 were shared.

Standard U-Net used ordinary encoder-decoder skip connections. U-Net+RES added the published BU-Net RES pathways but omitted WC. The full BU-Net reimplementaion combined RES with the published WC bottleneck module. RES used separable $N \times 1$ and $1 \times N$ branches with N in 9, 11, 13, and 15 across resolution levels, followed by fusion convolutions. WC used two oppositely ordered separable 15-pixel paths whose outputs were summed. Figure 3 identifies publication provenance directly.

The implementation followed the BU-Net prose where the original schematic was ambiguous. Deliberate implementation choices were four mutually exclusive output classes, base width 16, the stated dropout placement, and no imported external code. These choices make the present work a reimplementaion study, not an exact reproduction of the original paper.

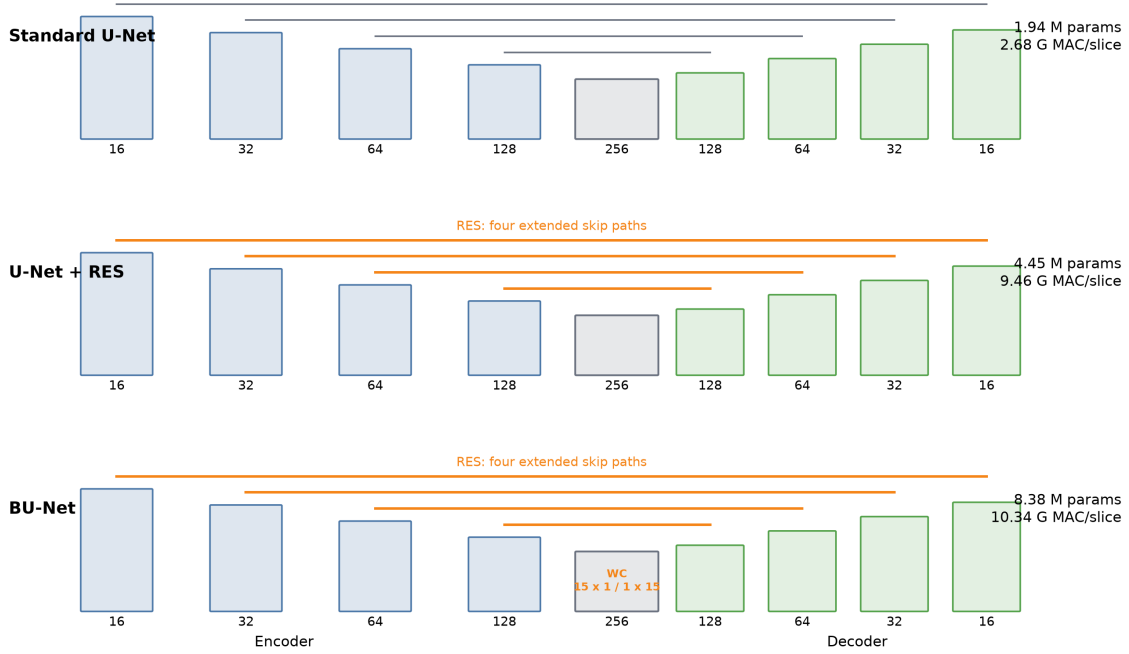


Figure 3: Compared 2D architectures. RES and WC are published BU-Net components from Rehman et al. [6].

3.5 Loss function and development screen

The selected loss was an equal-weight combination of channel-wise binary cross-entropy (BCE) and foreground focal Tversky loss (FTL):

$$\mathcal{L} = 0.5 \mathcal{L}_{BCE} + 0.5 \mathcal{L}_{FTL},$$

$$\mathcal{L}_{FTL} = \frac{1}{|C_f|} \sum_{c \in C_f} \left(1 - \frac{TP_c + \epsilon}{TP_c + \alpha FP_c + \beta FN_c + \epsilon} \right)^\gamma,$$

where C_f contains the three foreground classes, $\alpha = 0.3$, $\beta = 0.7$, $\gamma = 0.75$, and $\epsilon = 10^{-5}$. BCE used sigmoid probabilities and one-hot targets; FTL used softmax probabilities. Inference used argmax over the four output classes. No class weights were applied.

The development screen evaluated six architecture arms and seven loss arms once each on the validation subset. Its role was shortlisting, not hypothesis testing or reporting a generalization estimate.

Table 2. Single-seed architecture development screen (n=37 validation patients).

Architecture arm	Mean regional Dice	Eliminated
BU-Net	0.751	no
Residual-block U-Net	0.728	yes
Residual-block U-Net+WC	0.741	no
Standard U-Net	0.721	yes
U-Net+RES	0.745	no
U-Net+WC	0.741	no

Table 3. Single-seed loss development screen (n=37 validation patients).

Loss arm	Mean regional Dice	Eliminated
Cross-entropy + soft Dice	0.721	yes
Binary cross-entropy	0.623	yes
Binary cross-entropy + focal Tversky	0.744	no
Binary cross-entropy + soft Dice	0.711	yes
Cross-entropy	0.700	yes
Focal Tversky	0.245	yes
Soft Dice	0.238	yes

3.6 Matched training and multi-seed confirmation

Every reportable run used the same Apple M1 Max MPS device, inputs, batch size 16, AdamW optimizer, learning rate 0.001, weight decay 0.00001, augmentation, and loss. Mixed precision and pretraining were disabled. The realized bounded protocol stopped each run at 2,000 optimizer steps or 0.5 accelerator-hours, whichever occurred first. A 200-step linear warmup preceded cosine decay to 0.01 of the initial rate. Validation occurred at step 2,000, and the highest patient-level mean regional Dice checkpoint was retained.

Three seeds (3 per candidate) confirmed standard U-Net, BU-Net, U-Net+RES, and U-Net+WC. U-Net+WC was eliminated by the predeclared rule. BU-Net and U-Net+RES then received two additional seeds each. Five-seed validation means were 0.743 and 0.740, respectively; the U-Net+RES minus BU-Net paired difference was -0.003 (95% bootstrap CI -0.012 to +0.004). Because this interval included zero, the development ranking was not interpreted as superiority. Standard U-Net, BU-Net, and U-Net+RES were frozen for internal-test evaluation.

3.7 Guarded internal-test evaluation

Before test access, candidate identities, 13 checkpoint hashes, the split hashes, the endpoint, three paired contrasts, inference rules, and statistical procedures were frozen. Test evaluation required an explicit authorization flag and an append-only access log. The test manifest was opened once. No checkpoint, threshold, post-processing stage, or model-selection decision changed afterward.

For each candidate, per-seed endpoint values were averaged within patient before patient-level statistical inference. No test-time augmentation or post-processing was used.

3.8 Outcomes and metric rules

The primary outcome was each patient’s arithmetic mean of WT, TC, and ET Dice. Region Dice values were also reported separately. Secondary estimates included HD95, surface Dice at 1 mm, lesion recall, lesion precision, lesion-wise Dice, false-positive lesion count, and relative volume error.

Lesions used 26-connectivity with a one-voxel minimum. Predicted and reference lesions were paired by maximum-total-IoU matching. If both masks were empty, overlap and surface Dice were 1 and HD95 was 0 mm; if only one mask was empty, overlap and surface Dice were 0 and HD95 was infinity. Undefined rates remained missing. Tables state finite denominators and infinity counts where applicable.

3.9 Statistical analysis

The patient was the only inferential unit. For each candidate, 95% confidence intervals used 10,000 patient-level bootstrap resamples with seed 20260729. Three frozen paired comparisons of mean regional Dice were tested with 100,000 two-sided sign-flip permutations using seed 20260730. Holm’s sequential procedure controlled the family-wise error rate at 0.05 [13]. We report paired mean differences, bootstrap intervals, paired standardized effect d_z , raw p values, and Holm-adjusted p values. Regional and secondary endpoints were estimation-only; no unplanned multiplicity-adjusted claims were made.

Grade, reference ET presence, and training-derived WT burden tertiles were exploratory. The ET-absent group was descriptive because it contained only five patients.

3.10 Resource profiling and reproducibility

Parameter counts and multiply-accumulate operations (MACs) were computed from the implemented models at a 4 x 240 x 240 slice input. One MAC was reported as two floating-point operations (FLOPs). Per-volume inference latency and allocated memory were measured on the same Apple M1 Max host and summarized across seeds. Development accelerator-hours were retained from each run. These measurements compare the present implementations on one host; they are not deployment benchmarks.

All tables and figures were generated from machine-readable run artifacts. A tracked manifest records hashes for reportable files. A clean-clone audit rebuilt the Gate 12 outputs twice, confirmed byte-identical results, ran static checks and tests, and verified a clean worktree.

4. Results

4.1 Cohort integrity and development selection

The content audit identified complete imaging and labels for all 369 canonical patients, no file-integrity errors, and no patient overlap across partitions. The maximum absolute standardized mean difference across continuous split features was 0.048, below the frozen tolerance of 0.35.

In the single-seed screen, BU-Net had the highest architecture-screen mean regional Dice (0.751); U-Net+RES was within the practical screening margin. BCE+FTL had the highest loss-screen mean (0.744). These development observations only determined which arms advanced.

In the three-seed stage, mean regional Dice was 0.739 for standard U-Net, 0.745 for BU-Net, 0.739 for U-Net+RES, and 0.730 for U-Net+WC. Only U-Net+WC met the elimination rule. The finalist interval spanning zero justified carrying both BU-Net and U-Net+RES into the frozen test analysis.

4.2 Primary internal-test outcome

All 74 test patients had finite primary outcomes. Mean regional Dice was 0.736 for standard U-Net, 0.752 for BU-Net, and 0.756 for U-Net+RES.

Table 4. Internal held-out test Dice estimates (n=74 patients).

Candidate	Mean regional Dice	95% bootstrap CI	WT Dice	TC Dice	ET Dice
Standard U-Net	0.736	0.692-0.775	0.810	0.693	0.704
BU-Net	0.752	0.716-0.786	0.823	0.724	0.709
U-Net+RES	0.756	0.719-0.790	0.827	0.726	0.715

BU-Net exceeded standard U-Net by +0.017 (95% CI +0.005 to +0.030; d_z =+0.308; raw p=0.007; Holm p=0.014). U-Net+RES exceeded standard U-Net by +0.020 (95% CI +0.008 to +0.034; d_z =+0.347; raw p=0.001; Holm p=0.004).

BU-Net minus U-Net+RES was -0.004 (95% CI -0.008 to -0.000; d_z =-0.229; Holm p=0.050). The estimate was small and close to the multiplicity threshold; it should not be read as a broad ranking beyond this protocol.

Table 5. Frozen paired comparisons for the primary outcome.

Contrast	Paired difference	95% bootstrap CI	d_z	Raw p	Holm p
BU-Net - Standard U-Net	+0.017	+0.005 to +0.030	+0.308	0.007	0.014
U-Net+RES - Standard U-Net	+0.020	+0.008 to +0.034	+0.347	0.001	0.004
BU-Net - U-Net+RES	-0.004	-0.008 to -0.000	-0.229	0.050	0.050

4.3 Regional, boundary, and lesion estimates

The largest regional overlap differences relative to standard U-Net occurred for TC. ET overlap changed less. Surface Dice means favored U-Net+RES numerically across all three regions. In contrast, finite HD95 means were lower for standard U-Net than for either component-based model; several TC and ET observations were infinite because one mask was empty. Thus, overlap improvements did not imply uniformly better boundary outlier behavior.

Table 6. Selected secondary outcomes. HD95 means use finite observations; infinity counts are shown in parentheses.

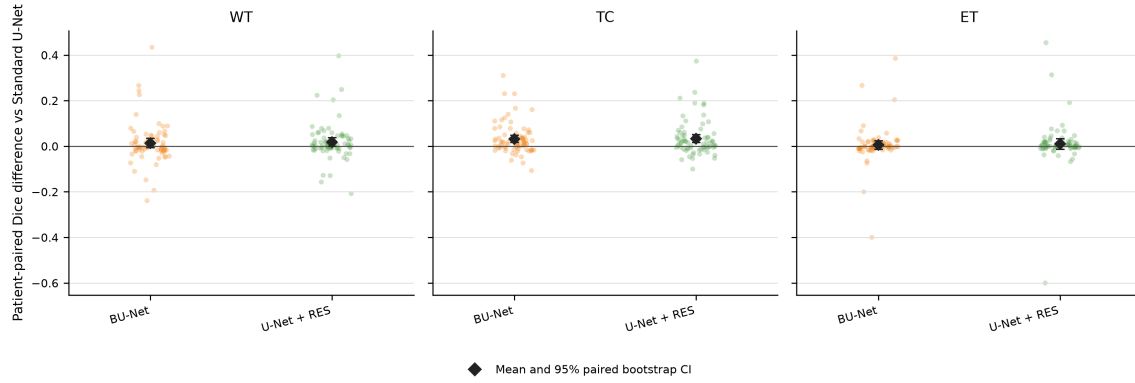


Figure 4: Patient-level paired regional Dice differences.

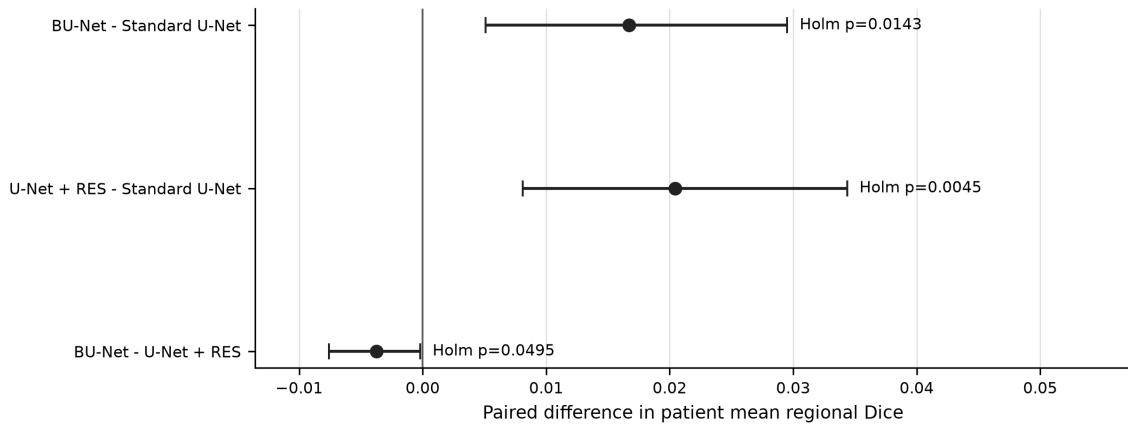


Figure 5: Frozen primary paired effects with 95% bootstrap confidence intervals.

	WT	TC	ET	WT	TC	ET		ET
	surface	surface	surface	HD95	HD95	HD95	ET lesion	lesion-
Candidate	Dice	Dice	Dice	mm	mm	mm	recall	wise Dice
Standard	0.605	0.577	0.734	21.9	17.4 (3	14.8 (4	0.470	0.356
U-Net					inf)	inf)		
BU-Net	0.621	0.586	0.744	35.8	22.0 (1	16.8 (5	0.475	0.367
					inf)	inf)		
U-	0.637	0.595	0.748	31.9	21.0 (2	16.9 (5	0.476	0.370
Net+RES					inf)	inf)		

ET lesion recall was approximately 0.47 for all candidates, and ET lesion-wise Dice remained below region-level ET Dice. These estimates expose a lesion-detection limitation that a voxel-overlap summary alone would obscure.

4.4 Resource demand

BU-Net had 4.32 times the parameters and 3.87 times the MACs of standard U-Net. U-Net+RES achieved the highest primary mean with 4.450 million parameters, compared with 8.385 million for BU-Net. Its mean p50 latency was 1.189 s/volume versus 1.258 s/volume, and its mean peak allocated memory was 88.9 MB versus 234.8 MB. Development accelerator-hours per run were similar and do not include preprocessing or study-level engineering time.

Table 7. Architecture and measured resource profile on one host.

	Parameters, MAC/slice, FLOP/slice, p50				p95	Peak	Development	Mean
Candidate	M	G	G	s/volume	s/volume	allocated	GPU-	regional
						MB	h/run	Dice
Standard	1.943	2.676	5.353	0.492	0.528	67.0	0.332	0.736
U-Net								
BU-Net	8.385	10.344	20.688	1.258	1.285	234.8	0.384	0.752
U-	4.450	9.459	18.919	1.189	1.215	88.9	0.388	0.756
Net+RES								

4.5 Exploratory subgroups and qualitative review

Exploratory subgroup estimates are reported without confirmatory p values. HGG patients had higher mean scores than LGG patients for all models. The ET-absent subset contained five patients and is descriptive only. Training-set WT tertiles defined small (at most 64,267 mm³), medium (64,267 to 123,163.3 mm³), and large burden groups.

Table 8. Exploratory mean regional Dice by frozen subgroup.

Subgroup	n	Standard			Interpretation
		U-Net	BU-Net	U-Net+RES	
Grade: HGG	59	0.806	0.808	0.812	exploratory
Grade: LGG	15	0.457	0.534	0.537	exploratory

Subgroup	n	Standard U-Net	BU-Net	U-Net+RES	Interpretation
ET reference: present	69	0.759	0.773	0.776	exploratory
ET reference: absent	5	0.418	0.470	0.480	descriptive_insufficient_n
WT burden: small	24	0.713	0.735	0.746	exploratory
WT burden: medium	24	0.770	0.771	0.772	exploratory
WT burden: large	26	0.724	0.751	0.750	exploratory

The qualitative panel includes preselected BU-Net success, hard, and failure cases: BraTS20_Training_166 (ET Dice 0.924), BraTS20_Training_137 (ET Dice 0.788), and BraTS20_Training_323 (ET Dice 0.000). The failure case demonstrates that a favorable cohort mean can coexist with near-zero ET overlap in an individual patient. Modality, prediction, false-positive, and false-negative panels were generated from retained test artifacts rather than selected after manuscript inspection.

5. Discussion

5.1 Principal findings

This controlled study supports three bounded conclusions. First, both BU-Net and U-Net+RES produced small patient-level improvements in mean regional Dice over standard U-Net after multiplicity correction. Second, U-Net+RES slightly exceeded the full BU-Net on the frozen primary endpoint. Third, BU-Net required substantially more parameters, MACs, latency, and allocated memory than U-Net+RES. Under this implementation and run budget, WC therefore did not provide an observable advantage over RES alone.

The effect sizes were small. BU-Net’s paired standardized effect versus standard U-Net was +0.308, and the BU-Net versus U-Net+RES contrast was -0.229. The latter p value was near 0.05 and its confidence interval lay close to zero. The evidence should be interpreted as a component-specific result under one bounded protocol, not a universal claim that WC is ineffective.

5.2 What the secondary outcomes add

Dice and surface Dice generally favored the component-based models, but finite HD95 means did not. Standard U-Net had lower finite HD95 means in WT, TC, and ET. This discordance can arise because Dice summarizes overlap while HD95 emphasizes distant boundary errors and becomes infinite when only one mask is empty. Reporting both prevents a one-dimensional account of segmentation quality.

Lesion-level estimates also temper the voxel-level results. ET lesion recall and lesion-wise Dice remained modest for every candidate. A model can achieve a high regional Dice on a large focus while missing small enhancing foci or generating disconnected false-positive components. The qualitative failure case reinforces this limitation.

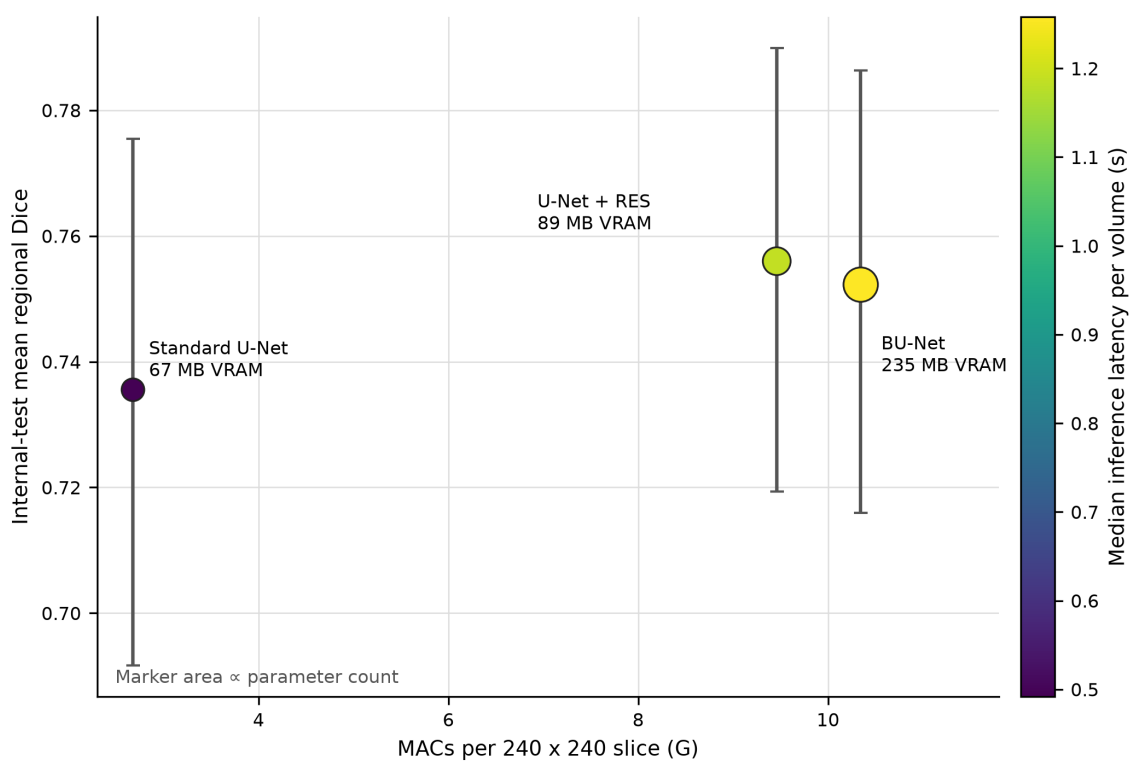


Figure 6: Mean regional Dice in relation to model and runtime resource measures.

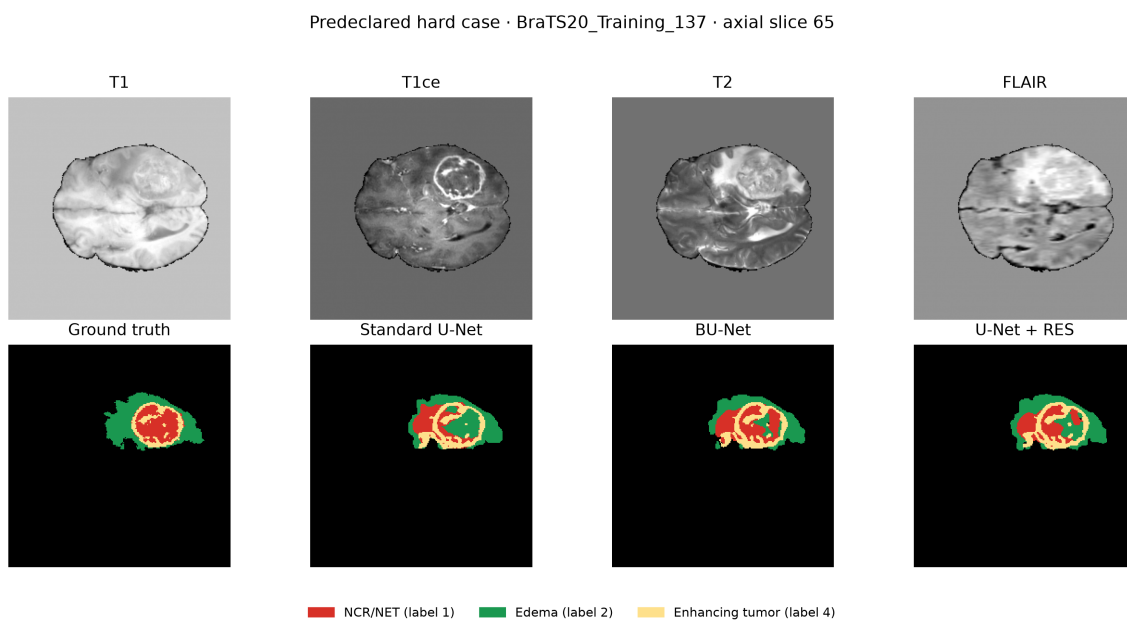


Figure 7: Multimodal MRI, reference labels, and frozen candidate predictions for an internal-test case.

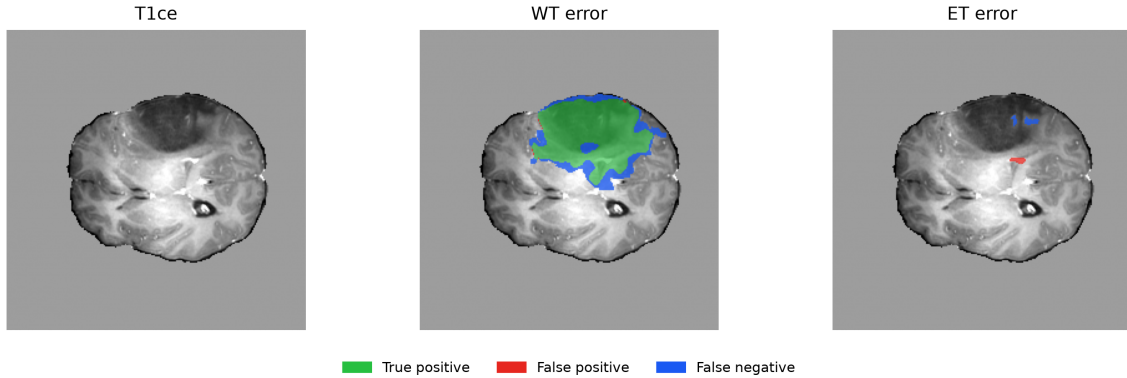


Figure 8: False-positive and false-negative error overlays.

5.3 Development ranking and test behavior

BU-Net ranked first during five-seed development, but U-Net+RES ranked first on the internal test. The development confidence interval between the finalists included zero, so freezing both was important. Selecting only the numerically leading validation model would have hidden this uncertainty. The result illustrates why seed replication, frozen finalist rules, and complete candidate reporting matter.

5.4 Scope relative to broader segmentation systems

nnU-Net, TransBTS, nnFormer, and other 3D systems are relevant modern comparators [9,14,15]. We did not train them. Nor did we perform cross-dataset or multi-institutional external validation. Literature scores cannot substitute for those experiments because acquisition, cohort, preprocessing, compute, and evaluation rules differ. Our results should therefore be read as a reproducible internal ablation of published BU-Net components, not as a leaderboard or state-of-the-art comparison.

5.5 Limitations

This study has several limitations. It used one public dataset and an internal split; external generalization is unknown. The networks were 2D and cannot use through-plane context as a 3D model can. Training was deliberately bounded at 2,000 optimizer steps or 0.5 accelerator-hours, so the results do not describe fully converged scaling behavior. The architecture widths and output encoding are reimplementations rather than a bit-exact reproduction of BU-Net. Resource measurements came from one Apple M1 Max host and do not establish deployment performance on clinical hardware. The 1 mm surface tolerance was predeclared but not calibrated against expert interobserver variability. HGG/LGG labels were inherited from the challenge release and were not independently adjudicated. Exploratory subgroups were small, especially the five-patient ET-absent subset. Finally, no prospective workflow, reader study, clinical endpoint, or regulatory assessment was performed.

5.6 Reproducibility and auditability

The released repository contains configuration files, split hashes, model and loss code, evaluator tests, analysis scripts, machine-readable outputs, figure and table generators, resource profiles, a claim ledger, and a tracked-artifact manifest. A clean-clone reproduction report records the exact commit and verifies deterministic

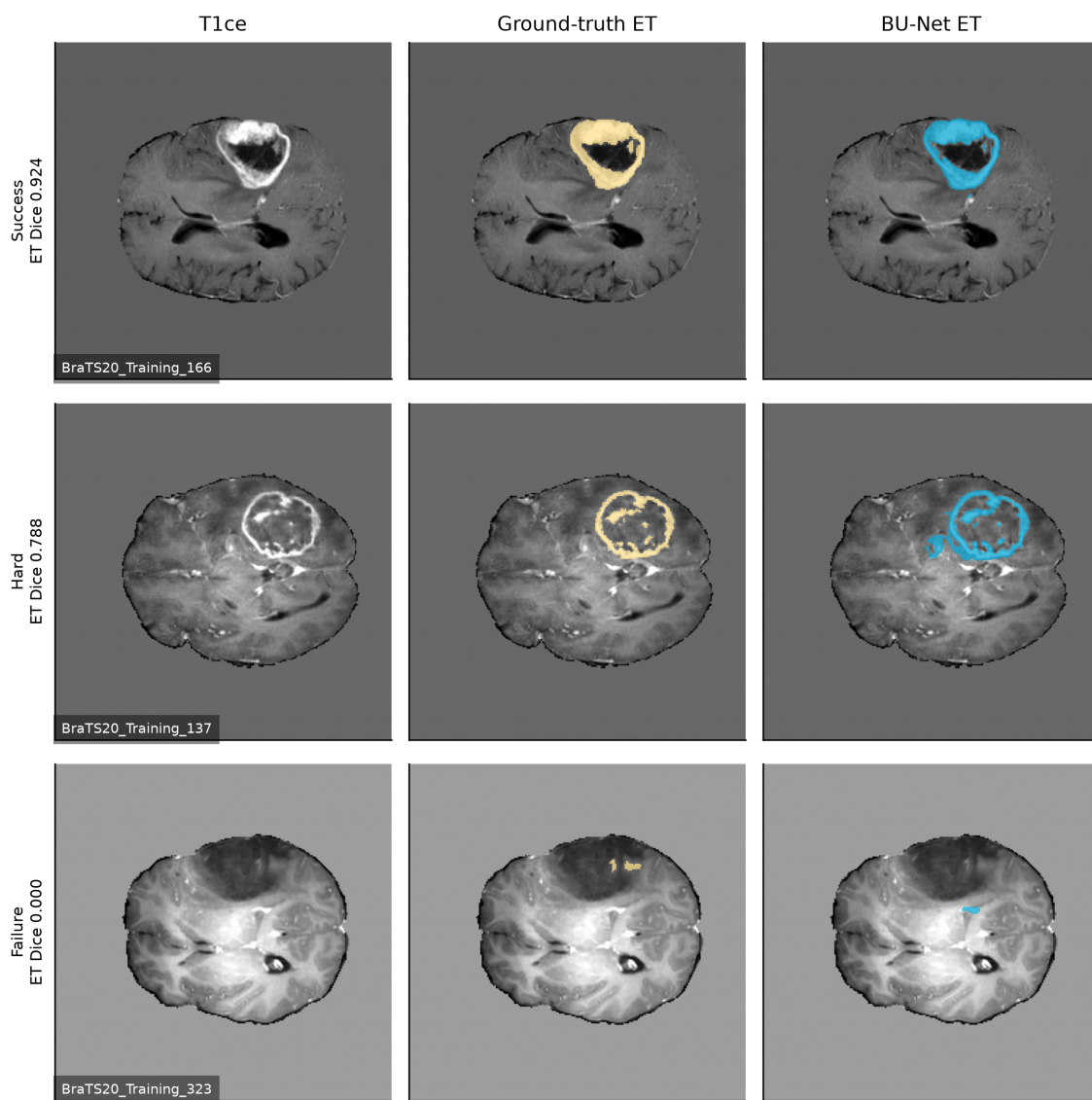
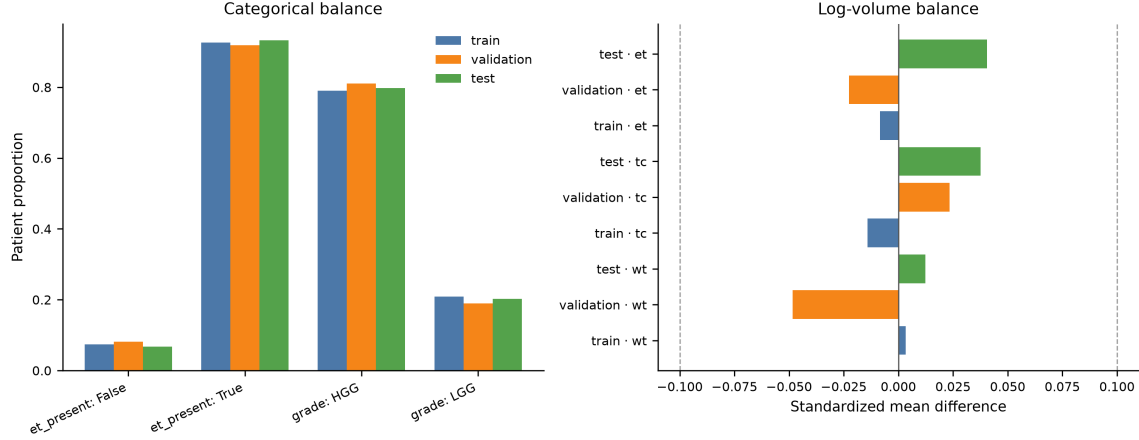


Figure 9: Prespecified BU-Net ET success, hard, and failure cases.



Supplementary Figure S1. Patient-level balance across frozen partitions.

output generation. Raw BraTS data and model checkpoints are excluded because redistribution is not authorized; users must obtain the dataset separately and configure local paths.

6. Conclusion

In a leakage-safe, patient-level, multi-seed evaluation of three matched 2D U-Net-family candidates, the published BU-Net RES component improved mean regional Dice relative to standard U-Net. Adding the published WC component in the full BU-Net did not improve the primary endpoint over RES alone and increased computational demand. Boundary and lesion-level results were more mixed than overlap scores. The work provides reproducible internal evidence about these components, while leaving 3D comparison, external validation, clinical calibration, and prospective utility as open questions.

Data and code availability

Code, frozen manifests, derived reports, tables, figures, and reproduction instructions are available at <https://github.com/TerekliTahaBerk/bratsarticle>. BraTS images and labels are not redistributed. Access to the original dataset is governed by the BraTS data providers. The internal held-out test partition is a study-specific subset of the BraTS 2020 training cohort and is not an official challenge test set.

Ethics statement

This computational study used a publicly distributed, de-identified challenge dataset and involved no direct participant contact or intervention. Authors should confirm the final journal-specific institutional review and exemption wording before submission.

Declarations requiring author confirmation

Funding, competing interests, author contributions, and institutional review wording were not inferable from the repository and must be confirmed by the authors in the target journal’s required format. No declaration has been invented in this manuscript package.

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